

6 (long)

A simply supported L-shaped elastic beam is shown in Fig. 6(a). The beam has second moment of area I and Young's modulus E . The beam is subjected to a horizontal point load of magnitude W in the middle of span AB and a vertical load of magnitude W distributed uniformly over span BC.

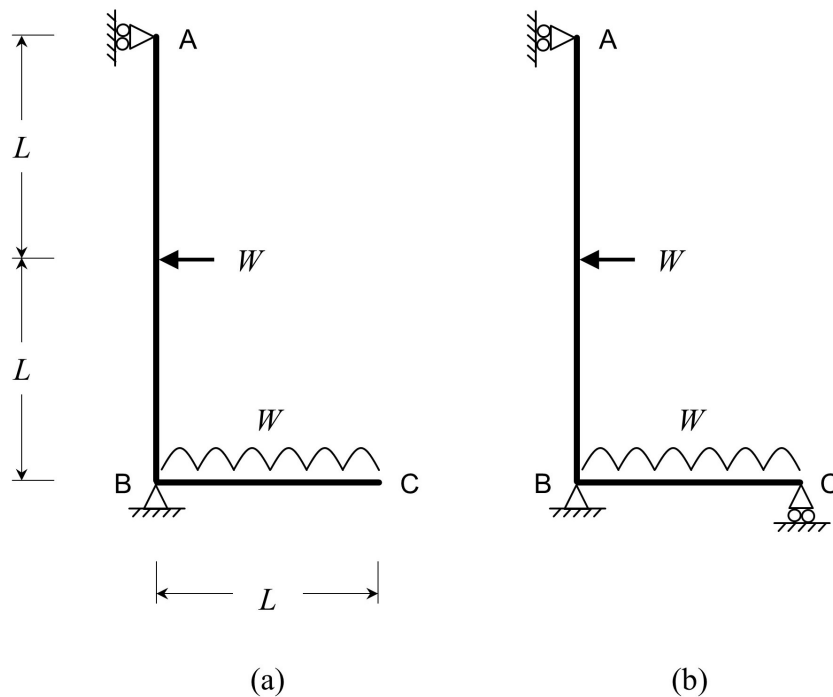


Fig. 6

- (a) Draw the shear and bending moment diagrams for the beam subject to this loading arrangement, noting all salient values. [10]
- (b) Determine the vertical deflection of point C. [10]
- (c) A roller support is added at point C as shown in Fig. 6(b), rendering the structure indeterminate. Determine the magnitude of the vertical reaction provided by the roller support at C. [10]